ENGINEERING MATHEMATICS

UNIT–III : DIFFERENTIAL CALCULUS AND APPLICATIONS

(As per DCET C20 Diploma Curriculum — Marks: 4)

DCET Exam Note:

Syllabus Coverage:

- Derivatives of continuous functions in an interval (List of formulae)
- Rules of differentiation
- Successive differentiation (up to second order)
- Applications of differentiation

1. BASIC CONCEPTS

Derivative Definition

The derivative of a function $f(x)$ at a point $x = a$ is defined as:

$$f'(a) = \lim_{h \to 0} \frac{f(a+h) - f(a)}{h}$$

If this limit exists, the function is differentiable at $x = a$.

Important:

Notation:
$f'(x)$ or $\dfrac{dy}{dx}$ or $Df(x)$ or y'
All mean the same thing: derivative of $f(x)$ with respect to x

2. STANDARD DERIVATIVES (MUST MEMORIZE)

Formula:

Basic Functions:

$\frac{d}{dx}(c) = 0$ (where c is constant)

$\frac{d}{dx}(x^n) = nx^{n-1}$ (Power Rule)

$\frac{d}{dx}(e^x) = e^x$

$\frac{d}{dx}(a^x) = a^x \ln a$ $(a > 0, a \neq 1)$

$\frac{d}{dx}(\ln x) = \frac{1}{x}$ $(x > 0)$

Formula:

Trigonometric Functions:

$\frac{d}{dx}(\sin x) = \cos x$

$\frac{d}{dx}(\cos x) = -\sin x$

$\frac{d}{dx}(\tan x) = \sec^2 x$

$\frac{d}{dx}(\cot x) = -\csc^2 x$

$\frac{d}{dx}(\sec x) = \sec x \tan x$

$\frac{d}{dx}(\csc x) = -\csc x \cot x$

Formula:

Inverse Trigonometric Functions:

$\frac{d}{dx}(\sin^{-1} x) = \frac{1}{\sqrt{1-x^2}}$ $(-1 < x < 1)$

$\frac{d}{dx}(\cos^{-1} x) = \frac{-1}{\sqrt{1-x^2}}$ $(-1 < x < 1)$

$\frac{d}{dx}(\tan^{-1} x) = \frac{1}{1+x^2}$

$\frac{d}{dx}(\cot^{-1} x) = \frac{-1}{1+x^2}$

3. RULES OF DIFFERENTIATION

3.1 Sum/Difference Rule

Formula:

f $y = u(x) \pm v(x)$, then:

$$\frac{dy}{dx} = \frac{du}{dx} \pm \frac{dv}{dx}$$

Example: If $y = x^3 + \sin x$, then $y' = 3x^2 + \cos x$

3.2 Product Rule

Formula:

f $y = u(x) \cdot v(x)$, then:

$$\frac{dy}{dx} = u\frac{dv}{dx} + v\frac{du}{dx}$$

Mnemonic: "First times derivative of second + second times derivative of first"

Illustration 3.1

Differentiate $y = x^2 \sin x$

Solution: Let $u = x^2$, $v = \sin x$

$u' = 2x$, $v' = \cos x$

$y' = uv' + vu' = x^2 \cos x + \sin x \cdot 2x = x^2 \cos x + 2x \sin x$

3.3 Quotient Rule

Formula:

f $y = \dfrac{u(x)}{v(x)}$, then:

$$\frac{dy}{dx} = \frac{v\frac{du}{dx} - u\frac{dv}{dx}}{v^2}$$

Mnemonic: "Low D-high minus high D-low over low squared"

Illustration 3.2

Differentiate $y = \dfrac{x}{x^2+1}$

Solution: Let $u = x$, $v = x^2 + 1$

$u' = 1$, $v' = 2x$

$$y' = \frac{vu' - uv'}{v^2} = \frac{(x^2+1)(1) - x(2x)}{(x^2+1)^2} = \frac{x^2+1-2x^2}{(x^2+1)^2} = \frac{1-x^2}{(x^2+1)^2}$$

3.4 Chain Rule

Formula:

f $y = f(g(x))$, then:

$$\frac{dy}{dx} = f'(g(x)) \cdot g'(x)$$

Alternative notation: If $y = f(u)$ and $u = g(x)$, then:

$$\frac{dy}{dx} = \frac{dy}{du} \cdot \frac{du}{dx}$$

Illustration 3.3

Differentiate $y = \sin(3x^2)$

Solution: Let $u = 3x^2$, then $y = \sin u$

$\dfrac{du}{dx} = 6x$, $\dfrac{dy}{du} = \cos u$

$\dfrac{dy}{dx} = \cos u \cdot 6x = 6x\cos(3x^2)$

4. SUCCESSIVE DIFFERENTIATION

Successive Derivatives

First derivative: $y' = \dfrac{dy}{dx} = f'(x)$

Second derivative: $y'' = \dfrac{d^2y}{dx^2} = f''(x)$

Third derivative: $y''' = \dfrac{d^3y}{dx^3} = f'''(x)$

n-th derivative: $y^{(n)} = \dfrac{d^ny}{dx^n} = f^{(n)}(x)$

4.1 Standard Patterns for Successive Differentiation

Formula:

For $y = x^n$:
$y' = nx^{n-1}$
$y'' = n(n-1)x^{n-2}$
$y''' = n(n-1)(n-2)x^{n-3}$
In general: $y^{(k)} = n(n-1)(n-2)\cdots(n-k+1)x^{n-k}$
For $k = n$: $y^{(n)} = n!$ (constant)

4.2 Trigonometric Functions

Formula:

For $y = \sin(ax+b)$:
$y' = a\cos(ax+b)$
$y'' = -a^2\sin(ax+b)$
$y''' = -a^3\cos(ax+b)$
$y^{(4)} = a^4\sin(ax+b)$
Pattern repeats every 4 derivatives

4.3 Exponential Functions

Formula:

For $y = e^{ax}$:
$y' = ae^{ax}$
$y'' = a^2e^{ax}$
$y''' = a^3e^{ax}$
In general: $y^{(n)} = a^ne^{ax}$

Illustration 4.1

Find second derivative of $y = x^3 + 3x^2 + 2x + 1$

Solution: $y' = 3x^2 + 6x + 2$
$y'' = 6x + 6$

Illustration 4.2

Find second derivative of $y = \sin 2x$

Solution: $y' = 2\cos 2x$
$y'' = -4\sin 2x$

5. APPLICATIONS OF DIFFERENTIATION

5.1 Tangent and Normal

Key Formulas

Slope of tangent at (x_1, y_1): $m_t = \left.\dfrac{dy}{dx}\right|_{(x_1,y_1)}$
Slope of normal at (x_1, y_1): $m_n = -\dfrac{1}{m_t}$ (if $m_t \neq 0$)
Equation of tangent: $y - y_1 = m_t(x - x_1)$
Equation of normal: $y - y_1 = m_n(x - x_1)$

Illustration 5.1

Find equation of tangent and normal to $y = x^2$ at $(2, 4)$

Solution: $\dfrac{dy}{dx} = 2x$
At $(2, 4)$: $m_t = 2(2) = 4$
$m_n = -\dfrac{1}{4}$
Tangent: $y - 4 = 4(x - 2) \Rightarrow y = 4x - 4$
Normal: $y - 4 = -\frac{1}{4}(x - 2) \Rightarrow 4y - 16 = -x + 2 \Rightarrow x + 4y = 18$

5.2 Rate of Change

Formula:

f $y = f(x)$, then $\dfrac{dy}{dx}$ represents the rate of change of y with respect to x.

Illustration 5.2

The radius of a circle is increasing at 3 cm/s. Find rate of change of area when radius = 5 cm.

Solution: $A = \pi r^2$
$\frac{dA}{dt} = 2\pi r \frac{dr}{dt}$
Given: $\frac{dr}{dt} = 3$ cm/s, $r = 5$ cm
$\frac{dA}{dt} = 2\pi(5)(3) = 30\pi$ cm²/s

5.3 Maxima and Minima

Steps to Find Local Extrema

1. Find $f'(x)$ and set $f'(x) = 0$ to get critical points
2. Find $f''(x)$
3. Evaluate $f''(x)$ at each critical point:
   - If $f''(x) > 0 \to$ **Local minimum**
   - If $f''(x) < 0 \to$ **Local maximum**
   - If $f''(x) = 0 \to$ Test fails, use first derivative test

Illustration 5.3

Find local maxima/minima of $f(x) = x^3 - 3x^2 + 4$

Solution: $f'(x) = 3x^2 - 6x = 3x(x-2)$
Set $f'(x) = 0$: $3x(x-2) = 0 \Rightarrow x = 0, 2$
$f''(x) = 6x - 6$
At $x = 0$: $f''(0) = -6 < 0 \to$ **Local maximum**
At $x = 2$: $f''(2) = 6 > 0 \to$ **Local minimum**
$f(0) = 4$, $f(2) = 0$
Local max at $(0, 4)$, **Local min at** $(2, 0)$

5.4 Increasing/Decreasing Functions

Formula:

or function $f(x)$:

- $f'(x) > 0 \to f(x)$ is **increasing**
- $f'(x) < 0 \to f(x)$ is **decreasing**
- $f'(x) = 0 \to$ Critical point (may be max/min)

6. SPECIAL TECHNIQUES

6.1 Logarithmic Differentiation

When to Use

Use when function is:

- Product of many functions: $y = f_1(x) \cdot f_2(x) \cdots f_n(x)$
- Power of functions: $y = [f(x)]^{g(x)}$
- Functions with variables in exponent: $y = x^x$

Illustration 6.1

Differentiate $y = x^x$

Solution: Take ln on both sides: $\ln y = x \ln x$

Differentiate: $\frac{1}{y}\frac{dy}{dx} = \ln x + x \cdot \frac{1}{x} = \ln x + 1$

$\frac{dy}{dx} = y(\ln x + 1) = x^x(\ln x + 1)$

6.2 Parametric Differentiation

Formula:

f $x = f(t)$ and $y = g(t)$, then:

$$\frac{dy}{dx} = \frac{dy/dt}{dx/dt} = \frac{g'(t)}{f'(t)}$$

Illustration 6.2

If $x = a\cos\theta$, $y = a\sin\theta$, find $\dfrac{dy}{dx}$

Solution: $\dfrac{dx}{d\theta} = -a\sin\theta$, $\dfrac{dy}{d\theta} = a\cos\theta$

$$\frac{dy}{dx} = \frac{a\cos\theta}{-a\sin\theta} = -\cot\theta$$

Practice Problems (DCET Pattern)

DCET Exam Note:

Solve these typical DCET questions:

1. Differentiate: $y = 3x^4 - 2x^3 + 5x - 7$ [Ans: $12x^3 - 6x^2 + 5$]
2. Find derivative: $y = \sqrt{x} + \dfrac{1}{\sqrt{x}}$ [Ans: $\frac{1}{2\sqrt{x}} - \frac{1}{2x^{3/2}}$]
3. Differentiate using product rule: $y = x^2 e^x$ [Ans: $x^2 e^x + 2xe^x$]
4. Differentiate using quotient rule: $y = \dfrac{\sin x}{x}$ [Ans: $\frac{x\cos x - \sin x}{x^2}$]
5. Find second derivative of $y = \cos 3x$ [Ans: $-9\cos 3x$]
6. Find slope of tangent to $y = x^2 - 4x$ at $x = 3$ [Ans: 2]
7. Find rate of change of area of circle w.r.t radius when $r = 7$ cm [Ans: 14π cm²/cm]
8. Find local maxima/minima of $f(x) = 2x^3 - 3x^2 - 12x + 5$ [Ans: Max at $x = -1$, Min at $x = 2$]
9. Differentiate using chain rule: $y = \ln(2x^2 + 1)$ [Ans: $\frac{4x}{2x^2+1}$]
10. If $x = t^2$, $y = t^3$, find $\dfrac{dy}{dx}$ [Ans: $\frac{3t}{2}$]

Quick Revision Points

- $\frac{d}{dx}(x^n) = nx^{n-1}$ (Power Rule)
- $\frac{d}{dx}(\sin x) = \cos x$
- $\frac{d}{dx}(\cos x) = -\sin x$
- Product Rule: $(uv)' = u'v + uv'$
- Quotient Rule: $(\frac{u}{v})' = \frac{u'v-uv'}{v^2}$
- Chain Rule: $\frac{dy}{dx} = \frac{dy}{du} \cdot \frac{du}{dx}$
- $y = e^{ax} \Rightarrow y^{(n)} = a^n e^{ax}$
- Tangent slope: $m_t = f'(x)$
- Normal slope: $m_n = -1/m_t$
- $f'(x) > 0 \rightarrow$ Increasing
- $f'(x) < 0 \rightarrow$ Decreasing
- $f''(x) > 0$ at critical point $\rightarrow$ Minimum
- $f''(x) < 0$ at critical point $\rightarrow$ Maximum

DCET Exam Note:

DCET Exam Tips:

- Memorize all standard derivatives
- Practice product/quotient/chain rules thoroughly
- For successive differentiation, remember patterns
- In applications: Tangent/Normal problems are common
- Maxima/Minima: Always check $f''(x)$ sign
- Rate problems: Use chain rule $\frac{dy}{dt} = \frac{dy}{dx} \cdot \frac{dx}{dt}$
- Logarithmic differentiation for complex functions
- Time management: 2-3 minutes per derivative calculation

Differential Calculus Formula Sheet

Standard Derivatives:

$$\frac{d}{dx}(x^n) = nx^{n-1}$$
$$\frac{d}{dx}(\sin x) = \cos x$$
$$\frac{d}{dx}(\cos x) = -\sin x$$
$$\frac{d}{dx}(e^x) = e^x$$
$$\frac{d}{dx}(\ln x) = \frac{1}{x}$$

Rules:

$$(u \pm v)' = u' \pm v'$$
$$(uv)' = u'v + uv'$$
$$\left(\frac{u}{v}\right)' = \frac{u'v - uv'}{v^2}$$
$$\frac{dy}{dx} = \frac{dy}{du} \cdot \frac{du}{dx}$$

Applications:

Tangent: $y - y_1 = m_t(x - x_1)$

Normal: $y - y_1 = m_n(x - x_1)$

$m_t = f'(x), \quad m_n = -\frac{1}{m_t}$

Max if $f''(x) < 0$

Min if $f''(x) > 0$